



DIFFERENTIAL EQUATIONS

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Partial differential equations

Basic concepts of partial differential equations

Partial differential equations (PDEs) arise in connection with various physical and geometrical problems when the function involved depend upon two or more independent variables. These variables may be the time t or more space coordinates x, y, z . Here, we give some basic concepts relating to PDEs, the formation and the methods for finding solutions of PDEs, and the method of separation of variables for the solutions of PDEs will also be discussed.

Definition:-

A partial differential equation is an equation involving partial derivatives of a function of two or more independent variables. A partial differential equation in the dependent variable u and independent variables x, y, z , and t is a relation of the form:

$$F(x, y, z, t; u, u_x, u_y, u_z, u_t, u_{xx}, u_{xy}, u_{xxx}, \dots) = 0.$$



Partial differential equations

For example:

$$3y^2 \frac{\partial u}{\partial x} + x^2 \frac{\partial u}{\partial y} = 2u, \quad (6)$$

$$\frac{\partial^2 u}{\partial x^2} + x^2 y \frac{\partial^2 u}{\partial y^2} = 0. \quad (7)$$

Where, u is the dependent variable, while x and y are the independent variables.

Order and degree of a partial differential equation

The order of a PDE is the order of the highest partial derivative that appears in the given equation and the greatest exponent of the highest partial derivative is called the degree of the given PDE. For example, equations (6) and (7) are of first and second-order, respectively, and both are of degree 1.



Partial differential equations

Whereas, the equation

$$\left(\frac{\partial^3 u}{\partial x^3}\right)^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)^3 + 8u = 0 \quad (8)$$

is of order 3 and degree 2.

Linear and non-linear partial differential equations

A partial differential equation is said to be linear if

- 1 it is of the first degree in the dependent variable and all of its partial derivatives;
- 2 the coefficients (of the dependent variable and its derivatives) are functions only of the independent variables.

Whereas, a PDE that is not linear is called as non-linear PDE. For example, equations (6) and (7) are linear, while the equation (8) is non-linear.



Partial differential equations

Homogeneous and non-homogeneous partial differential equations

A linear PDE is said to be homogeneous if each term in the equation contains either the dependent variable or one of its derivatives. Otherwise, it is said to be non-homogeneous or inhomogeneous PDE. For example, equations (6) and (7) are homogeneous, while the equation

$$\frac{\partial^2 u}{\partial x^2} + x^2 y \frac{\partial^2 u}{\partial y^2} = 2xy$$

is non-homogeneous.

General and particular solutions

A solution of a PDE is any function which satisfies the equation identically. The general solution is a solution which contains the number of arbitrary independent functions equal to the order of the equation. Whereas, a particular solution is one which can be obtained from the general solution by particular choice of arbitrary functions.



Partial differential equations

For example, as can be observed by substitution

$$u = x^2y - \frac{1}{2}xy^2 + F(x) + G(y)$$

is a solution of the PDE

$$\frac{\partial^2 u}{\partial x \partial y} = 2x - y.$$

Since it contains two arbitrary independent functions $F(x)$ and $G(y)$, so it is called as general solution. If, in-particular, $F(x) = 2 \sin x$ and $G(y) = 3y^4 - 5$ then we obtain the following particular solution

$$u = x^2y - \frac{1}{2}xy^2 + 2 \sin x + 3y^4 - 5.$$

Exercise:-

Exercise #12.1, Page # 542 (Ref-book no. 1).



Partial differential equations

Auxiliary conditions

Generally , the PDEs that represent physical systems have infinite number of solutions. For example, the functions:

$$u = x^2 - y^2, \quad u = e^x \cos y, \quad u = \log(x^2 + y^2), \quad u = \tan^{-1}\left(\frac{y}{x}\right), \dots$$

which are entirely different from each other are solutions of the following PDE

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

In order to obtain a unique solution of the PDE corresponding to a given physical problem, one must use additional information (i.e., auxiliary conditions) arising from the physical situation. They fall in two categories: **(i) Boundary conditions** **(ii) Initial conditions.**



Partial differential equations

Boundary conditions

These are the conditions that must be satisfied at points on the boundary S of the region R in which the PDE holds. Here, S may be the bounding curve of a plane region R or the surface of a three-dimensional region R . There are three types of boundary conditions:

- 1 Dirichlet condition

$$u = g \quad \text{on } S$$

- 2 Neumann (or flux) condition

$$\frac{\partial u}{\partial n} = g \quad \text{on } S$$

- 3 Mixed (or Robin or radiation) condition

$$\alpha u + \beta \frac{\partial u}{\partial n} = g \quad \text{on } S$$



Partial differential equations

where n denotes the (typically exterior) normal to the boundary and g, α, β are functions prescribed on S .

Note it that, the boundary must be sufficiently smooth in such a way that the normal derivative can exist, since, for example, at corner points on the boundary the normal vector is not well defined.

Initial conditions

These are the conditions that must be satisfied throughout the region R at the instant when consideration of the physical system begins. When time t is one of the independent variable and we specify a condition at $t = 0$, we refer to it as initial condition. A typical initial condition is said to be of Cauchy type if the values of both u and $\frac{\partial u}{\partial t}$ on the boundary at $t = 0$ (i.e., the initial values of u and $\frac{\partial u}{\partial t}$) are given.



Partial differential equations

Formation of PDEs

Partial differential equations may be derived either by the elimination of arbitrary constants from the given relation between the variables or by the elimination of arbitrary functions involving the variables.

Elimination of arbitrary constants

Case (1): If the number of arbitrary constants is equal to the number of independent variables, then we proceed as follows:

Let u be a function of two independent variables x and y defined by

$$f(x, y, u, a, b) = 0, \quad (9)$$

where a and b are two arbitrary constants. Differentiating equation (9) partially w.r.t. x and y , we get



Partial differential equations

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} = 0, \quad (10)$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} = 0. \quad (11)$$

Generally, the arbitrary constants may be eliminated from equations (9), (10), (11) yielding a PDE of order one

$$g\left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right) = 0.$$

Example:-

Form the PDEs by eliminating the arbitrary constants from the following relations:

$$i) \ u = ax + by, \quad ii) \ u = (x^2 + a)(y^2 + b).$$



Partial differential equations

Solution:- Take

$$u = ax + by. \quad (12)$$

Differentiating equation (12) partially w.r.t. x and y , we get

$$\frac{\partial u}{\partial x} = a, \quad (13)$$

$$\frac{\partial u}{\partial y} = b. \quad (14)$$

Substituting the values of a and b from equations (13) and (14) in equation (12), we get

$$u = x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$$

as the required partial differential equation.



Partial differential equations

Case (2): If u is a function of x and y defined by the relation involving only one arbitrary constant, then it is usually possible to obtain two distinct PDEs of order one by eliminating the constant.

Example:- Form the PDEs by eliminating the arbitrary constant a from the following relation

$$u = a(x + y).$$

Solution:- Take

$$u = a(x + y). \quad (15)$$

Differentiating equation (15) partially w.r.t. x , we get

$$\frac{\partial u}{\partial x} = a,$$

so that the following PDE is obtained

$$u = (x + y) \frac{\partial u}{\partial x}.$$



Partial differential equations

Similarly, differentiating equation (15) partially w.r.t. y , we get

$$\frac{\partial u}{\partial y} = a,$$

so that the PDE

$$u = (x + y) \frac{\partial u}{\partial y}$$

is obtained.

Case (3): If the number of arbitrary constants to be eliminated exceeds the number of independent variables, the resulting PDE is usually of higher order than the first. However, this PDE may not be unique.

Example:-

Form the PDEs by eliminating the arbitrary constants a, b, c from the following relation:

$$u = ax + by + cxy.$$



Partial differential equations

Solution:- Take

$$u = ax + by + cxy. \quad (16)$$

Differentiating equation (16) partially w.r.t. x and y , we get

$$\frac{\partial u}{\partial x} = a + cy, \quad (17)$$

$$\frac{\partial u}{\partial y} = b + cx. \quad (18)$$

These together with the given relation are not sufficient to eliminate 3 arbitrary constants. Differentiating equation (17) partially w.r.t. x , we have

$$\frac{\partial^2 u}{\partial x^2} = 0$$

a PDE order order 2.



Partial differential equations

Differentiating equation (18) partially w.r.t. y , we have

$$\frac{\partial^2 u}{\partial y^2} = 0$$

a PDE order order 2.

Differentiating equation (17) partially w.r.t. y or equation (18) partially w.r.t. x , we have

$$\frac{\partial^2 u}{\partial x \partial y} = c$$

From equation(17), we have

$$\frac{\partial u}{\partial x} = a + y \frac{\partial^2 u}{\partial x \partial y},$$

or $a = \frac{\partial u}{\partial x} - y \frac{\partial^2 u}{\partial x \partial y}.$



Partial differential equations

Similarly, from equation(18), we have

$$\frac{\partial u}{\partial y} = b + x \frac{\partial^2 u}{\partial x \partial y},$$
$$\text{or } b = \frac{\partial u}{\partial y} - x \frac{\partial^2 u}{\partial x \partial y}.$$

Substituting the values of a, b and c in the given relation, we get

$$u = \left(\frac{\partial u}{\partial x} - y \frac{\partial^2 u}{\partial x \partial y} \right) x + \left(\frac{\partial u}{\partial y} - x \frac{\partial^2 u}{\partial x \partial y} \right) y + xy \frac{\partial^2 u}{\partial x \partial y},$$
$$= x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} - xy \frac{\partial^2 u}{\partial x \partial y},$$

which is a partial differential equation of order 2. Thus, we have three PDEs of the same order associated with the given relation:

$$\frac{\partial^2 u}{\partial x^2} = 0, \quad \frac{\partial^2 u}{\partial y^2} = 0, \quad u = x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} - xy \frac{\partial^2 u}{\partial x \partial y}.$$



Partial differential equations

Example:- Obtain the PDEs by eliminating the arbitrary functions from the following relations:

$$i) \quad u = f\left(\frac{y}{x}\right) \quad ii) \quad u = F(x - ct) + G(x + ct)$$

Solution:- i) Take

$$u = f\left(\frac{y}{x}\right) \quad (19)$$

Differentiating equation (19) partially w.r.t. x and y , we get

$$\frac{\partial u}{\partial x} = -\frac{y}{x^2} f' \left(\frac{y}{x} \right) \quad (20)$$

$$\frac{\partial u}{\partial y} = \frac{1}{x} f' \left(\frac{y}{x} \right) \quad (21)$$

Multiplying equation (20) by x and equation (21) by y and then adding, we get



Partial differential equations

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0 \quad (22)$$

as the required partial differential equation of order one.

ii) Take

$$u = F(x - ct) + G(x + ct) \quad (23)$$

Differentiating twice equation (23) partially w.r.t. x , we get

$$\begin{aligned} \frac{\partial u}{\partial x} &= F'(x - ct) + G'(x + ct), \\ \frac{\partial^2 u}{\partial x^2} &= F''(x - ct) + G''(x + ct). \end{aligned} \quad (24)$$

Differentiating twice equation (23) partially w.r.t. t , we get

$$\frac{\partial u}{\partial t} = -cF'(x - ct) + cG'(x + ct)$$



Partial differential equations

$$\frac{\partial^2 u}{\partial t^2} = c^2 F''(x - ct) + c^2 G''(x + ct) \quad (25)$$

From equations (24) and (25), we have

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

a partial differential equation of order two.

Remarks:- The above equation is called as **one-dimensional wave equation** describing the transverse vibrations of a stretched string. The most generalised solution of the above equation is the equation (23). From these examples (as illustrated above) it is clear that a PDE can result both from the elimination of arbitrary constants and from the elimination of arbitrary functions.



Partial differential equations

Solutions of partial differential equations

If a partial differential equation involves derivatives w.r.t. one of the independent variables only, we can solve it like an ordinary differential equation treating the other independent variables as parameters. We explain the method with the following examples.

Example:- Find the solution of the following PDEs:

$$i) \ u_{xx} = 0 \quad ii) \ u_y = 2xyu.$$

Solution:- i) Take

$$u_{xx} = 0. \tag{26}$$

Integrating equation (26) w.r.t. x , we get

$$u_x = f(y)$$

where $f(y)$ is an arbitrary function. Integrating the above equation again w.r.t. x , we get



Partial differential equations

$$u(x, y) = xf(y) + g(y)$$

where $g(y)$ is an arbitrary function.

Solution:- ii) Take

$$u_y = 2xyu. \quad (27)$$

The above equation can be written as

$$\frac{u_y}{u} = 2xy.$$

Integrating the above equation w.r.t. y , we get

$$\begin{aligned} \int \frac{u_y}{u} dy &= \int 2xy dy \\ \ln u &= xy^2 + \ln f(x) \end{aligned}$$



Partial differential equations

where $\ln f(x)$ is an arbitrary function. Taking antilog of both sides of the above equation, we get

$$u(x, y) = f(x)e^{xy^2}.$$

Example:-

Solve the following partial differential equation

$$u_{xx} + 3u_x - 4u = 12.$$

Solution:- Let $D = \frac{\partial}{\partial x}$ and $D^2 = \frac{\partial^2}{\partial x^2}$, then the above equation becomes

$$(D^2 + 3D - 4)u = 12$$

Exercise:-

Exercise #12.1, Page # 543 (Ref-book no. 1).



Partial differential equations

Partial differential equations of order one

A partial differential equation of order one in the dependent variable u and independent variables x and y is a relation of the form

$$F(x, y, u, u_x, u_y) = 0.$$

Thus the general linear partial differential equation of order 1 is

$$au_x + bu_y + cu = d$$

where a, b, c , and d may be constants or may dependent on x and y but not on u . For example, the following are linear partial differential equations of order 1:

$$i) \quad u_x = 2xyu$$

$$ii) \quad u_x + u_y = 0$$

$$iii) \quad yu_x - xu_y = 0$$

$$iv) \quad xu_x + yu_y = 2(x + y)u$$



Partial differential equations

Partial differential equations of order two

A partial differential equation of order two in the dependent variable u and independent variables x and y is a relation of the form

$$F(x, y, u, u_x, u_y, u_{xx}, u_{xy}, u_{yy}) = 0.$$

Thus the general linear partial differential equation of order 2 is

$$au_{xx} + 2bu_{xy} + cu_{yy} + du_x + eu_y + fu = g$$

where a, b, c, d, e, f , and g may be constants or may dependent on x and y but not on u . If $g = 0$, then the equation is called **homogeneous**, while if $g \neq 0$, then it is called **non-homogeneous**. For example, the following are linear partial differential equations of order 2:

$$i) \quad u_{xx} = \frac{1}{c^2} u_{tt} \quad \text{One-dimensional wave equation}$$



Partial differential equations

$$ii) \quad u_{xx} = \frac{1}{c^2} u_t$$

One-dimensional heat equation

$$iii) \quad u_{xx} + u_{yy} = 0$$

Two-dimensional Laplace equation

$$iv) \quad u_{xx} + u_{yy} = f(x, y)$$

Two-dimensional Poisson equation

$$v) \quad u_{xx} + u_{yy} + u_{zz} = 0$$

Three-dimensional Laplace equation

Here, c^2 is a positive constant, t is time, and x, y, z are Cartesian coordinates. The equation (iv) with $f(x, y) \neq 0$ is non-homogeneous, whereas all the other equations are homogeneous.



Partial differential equations

Classification of second order partial differential equations

Consider the following general linear homogeneous partial differential equation of order two

$$au_{xx} + 2bu_{xy} + cu_{yy} + du_x + eu_y + fu = 0. \quad (28)$$

We note that the form of equation (28) resembles that of a general conic section

$$ax^2 + 2bxy + cy^2 + dx + ey + f = 0. \quad (29)$$

The above equation represents an ellipse, parabola, or hyperbola when the discriminant $b^2 - ac$ is negative, zero, or positive, respectively. We use the similar classification for the PDE (28) and say that it is elliptic, parabolic, or hyperbolic when the discriminant $b^2 - ac$ is negative, zero, or positive, respectively. Note it that, the type of equations (28) and (29) is determined solely by its principal part, i.e., the terms involving the highest-order derivatives of u and that the type will generally change with position in the xy -plane unless a, b, c are constants.



Partial differential equations

Example:- Classify the following partial differential equations:

$$i) \quad u_{xx} + u_{yy} = 0$$

$$ii) \quad u_{xx} = \frac{1}{\alpha^2} u_t$$

$$iii) \quad u_{xx} = \frac{1}{\alpha^2} u_{tt}$$

$$iv) \quad yu_{xx} + xu_{yy} = 0$$

Solution:- i) The two-dimensional Laplace's equation $u_{xx} + u_{yy} = 0$ may be obtained from equation (28) by putting $a = 1$, $b = 0$, $c = 1$, and $d = e = f = 0$. Since $b^2 - ac = -1 < 0$, therefore, the given PDE is of elliptic type.

Note it that, a PDE may be elliptic in one region of the plane, parabolic in another region, and hyperbolic in yet another region.



Partial differential equations

Method of separation of variables

In this method, it is assumed that the solution, i.e., dependent variable can be expressed as a product of unknown functions each of which depends upon only one of the independent variables. With this solution, the partial differential equation can be written in such a way that one side depends upon only one variable while the other side depends upon the remaining variables from which it is concluded that each side must be a constant.

By the repetition of this, the given PDE reduces to two or more ODEs each one involving one of the independent variable. Solving these ODEs, the unknown functions can be determined. The principle of superposition can then be used to find the actual solution of the given PDE. We illustrate this method of separation of variables with the following examples.



Partial differential equations

Example:- Using the method of separation of variables, solve the following partial differential equations:

$$i) \quad u_x + u_y = 0 \qquad ii) \quad u_{xy} = u.$$

Solution:- i) $u_x + u_y = 0$.

Let

$$u(x, y) = X(x)Y(y) = XY. \qquad (30)$$

Then by differentiation, we get $u_x = X'Y$ and $u_y = XY'$ so that the given partial differential equation becomes

$$X'Y + XY' = 0, \quad \text{where } X' = \frac{dX}{dx} \text{ and } Y' = \frac{dY}{dy}.$$

Separating variables, we get

$$\frac{X'}{X} = -\frac{Y'}{Y}.$$



Partial differential equations

The L.H.S. of the above equation depends upon x while the R.H.S. depends upon y only. Moreover, since they are equal, their common value cannot be a function of either x or y and must therefore be a constant, say, c , i.e.,

$$\frac{X'}{X} = -\frac{Y'}{Y} = c.$$

This implies two ordinary differential equations:

$$X' - cX = 0 \quad \text{and} \quad Y' + cY = 0,$$

whose solutions are: $X = Ae^{cx}$ and $Y = Be^{-cy}$, respectively. Where, A and B are arbitrary constants. Thus from equation (30), the general solution of the given partial differential equation is

$$u(x, y) = X(x)Y(y) = XY = Ae^{cx}Be^{-cy} = Ke^{c(x-y)}, \quad \text{where } K = AB.$$



Partial differential equations

Solution:- ii) $u_{xy} = u$.

Let

$$u(x, y) = X(x)Y(y) = XY. \quad (31)$$

Then by differentiation, we get $u_{xy} = X'Y'$ so that the given partial differential equation becomes

$$X'Y' = XY, \quad \text{where } X' = \frac{dX}{dx} \text{ and } Y' = \frac{dY}{dy}.$$

Separating variables, we get

$$\frac{X'}{X} = \frac{Y}{Y'}.$$



Partial differential equations

Example:-

Using the method of separation of variables, solve the following boundary value problems (BVPs):

$$i) \quad u_x = 4u_y \quad u(0, y) = 8e^{-3y}$$

$$ii) \quad u_x = 4u_y \quad u(0, y) = 8e^{-3y} + 4e^{-5y}$$

Solution:- i) $u_x = 4u_y$.

Let

$$u(x, y) = X(x)Y(y) = XY. \quad (32)$$

Then by differentiation, we get $u_x = X'Y$ and $u_y = XY'$ so that the given partial differential equation becomes

$$X'Y = 4XY', \quad \text{where } X' = \frac{dX}{dx} \text{ and } Y' = \frac{dY}{dy}.$$



Partial differential equations

Separating variables, we get

$$\frac{X'}{4X} = \frac{Y'}{Y}.$$

The L.H.S. of the above equation depends upon x while the R.H.S. depends upon y only. Moreover, since they are equal, their common value cannot be a function of either x or y and must therefore be a constant, say, c , i.e.,

$$\frac{X'}{4X} = -\frac{Y'}{Y} = c.$$

This implies two ordinary differential equations:

$$X' - 4cX = 0 \quad \text{and} \quad Y' - cY = 0,$$

whose solutions are: $X = Ae^{4cx}$ and $Y = Be^{cy}$, respectively. Where, A and B are arbitrary constants. From equation (32), the general solution of the given partial differential equation is



Partial differential equations

$$u(x, y) = X(x)Y(y) = XY = Ae^{4cx}Be^{cy} = Ke^{c(4x+y)}, \quad \text{where } K = AB.$$

Using the boundary condition $u(0, y) = 8e^{-3y}$, we have

$$8e^{-3y} = Ke^{cy}$$

which is possible if and if $K = 8$ and $c = -3$. Thus, from equation (32), we have

$$u(x, y) = 8e^{-3(4x+y)},$$

is the required solution.



Partial differential equations

Example:-

Obtain the partial differential equations by eliminating arbitrary constants from the following relations:

$$\begin{array}{ll} i) & u = ax + (1 - a)y + b \\ ii) & 2u = \frac{x^2}{a^2} + \frac{y^2}{b^2} \\ iii) & u = xy + y\sqrt{x^2 - a^2} + b \\ iv) & u = ae^{by} \sin bx \end{array}$$

Example:-

Obtain the partial differential equations by eliminating arbitrary functions from the following relations:

$$\begin{array}{ll} i) & u = y^2 + 2f\left(\frac{1}{x} + \ln y\right) \\ ii) & u = x^n f\left(\frac{y}{x}\right) \\ iii) & u = yf(x) + xg(y) \\ iv) & u = f(x - 3y) + g(2x + y) \end{array}$$



Partial differential equations

Example:-

Show that $u(x, y) = xf(2x + y)$ is a general solution of

$$x \frac{\partial u}{\partial x} - 2x \frac{\partial u}{\partial y} = u.$$

Then find a particular solution satisfying $u(1, y) = y^2$.

Example:-

Show that $u(x, t) = e^{-8t} \sin 2x$ is a solution to the following initial-boundary value problem:

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}$$

$$(B. C's) \quad u(0, t) = 0, \quad u(\pi, t) = 0, \quad t > 0$$

$$(I. C.) \quad u(x, 0) = \sin 2x$$



Partial differential equations

Example:-

Solve the following partial differential equation

$$\frac{\partial^3 u}{\partial y \partial x^2} = \cos(2x + 3y).$$

Example:-

Classify the following partial differential equations according to the type:

i) $3u_{xx} + 2u_{xy} + 5u_{yy} + xu_y = 0$

ii) $u_{xx} + 2yu_{xy} + xu_{yy} - u_x + u = 0$

iii) $2xyu_{xy} + xu_y + yu_x = 0$

iv) $yu_{xx} + 2xu_{xy} + yu_{yy} = 0$



Partial differential equations

Example:-

Using the method of separation of variables, solve the following partial differential equations:

$$i) \quad u_x + u_y = 2(x + y)u \qquad ii) \quad x^2 u_{xy} + 3y^2 u = 0$$

Example:-

Using the method of separation of variables, solve the following boundary value problems (BVPs):

$$\begin{aligned} i) \quad 3u_x + 2u_y &= 0, & u(x, 0) &= 4e^{-x} \\ ii) \quad u_x &= 2u_y + u, & u(x, 0) &= 3e^{-5x} + 2e^{-3x} \end{aligned}$$